

Portfolio Flood Risk Assessment: Process Overview

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1 Process Architecture

The portfolio flood risk assessment system consists of two primary components:

1. Property Valuation Pipeline
2. Flood Risk Assessment Pipeline

2 Data Flow

- Initial property portfolio data ingestion
- Property valuation and feature engineering
- Risk factor calculation and spatial analysis
- Portfolio-level flood impact assessment
- Results aggregation and reporting

3 Key Processes

1. Portfolio Data Processing
 - Data validation and cleaning
 - Geographic coordinate processing
 - Property characteristic normalization
2. Valuation Model Application
 - Feature extraction and transformation
 - Model prediction execution
 - Valuation adjustment calculations

3. Flood Risk Integration

- Spatial correlation analysis
- Flood depth calculations
- Impact assessment computation

4 Process Integration

The system integrates property valuation outputs with flood risk assessment through:

- Shared spatial indexing structures
- Unified data formats
- Synchronized calculation pipelines